

The $\mathfrak{m}_3$ Doublet and Custodial $\rho = 1$ in the $\mathbb{CP}^2$ /Aloff–Wallach Kaluza–Klein Carrier

KK Orchestra working lecture

June 1, 2026

Abstract

This lecture isolates the clean custodial part of the $\mathbb{CP}^2$ /Aloff–Wallach programme. The internal seven-manifold

$$SO(3) \longrightarrow X_{1,1} = SU(3)/U(1)_{1,1} \longrightarrow \mathbb{CP}^2$$

has a tangent split

$$\mathfrak{m} = \mathfrak{m}_0 \oplus \mathfrak{m}_3, \quad \dim_{\mathbb{R}} \mathfrak{m}_0 = 3, \quad \dim_{\mathbb{R}} \mathfrak{m}_3 = 4.$$

The four-dimensional block $\mathfrak{m}_3 \simeq \mathbb{C}^2 \simeq \mathbb{R}^4$ supplies the electroweak Higgs doublet. Since $\mathfrak{m}_3$ is a single irreducible real $U(2)$ -module, an invariant quadratic form restricts to one scalar on $\mathfrak{m}_3$. A vacuum in this single doublet block gives $\kappa_c = \kappa_n$ and hence the tree-level custodial relation $\rho = 1$. The vertical $\mathfrak{m}_0$ triplet, D6 branch terms, and De Vries spectral questions are separate consistency layers.

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1 Reading Map

The lecture has two audiences in mind.

- For a physicist, the central calculation is the electroweak mass matrix of one Higgs doublet, written with the internal metric factor carried along.
- For a geometer, the central point is the isotropy representation of $X_{1,1}$. The block $\mathfrak{m}_3$ is a real irreducible four-plane, while $\mathfrak{m}_0$ is the vertical three-plane.

The conclusion is deliberately modest. The $\rho = 1$ result is a standard custodial statement with a geometric origin. It does not derive a De Vries numerical relation. De Vries remains a separate source-action problem involving a named operator, a retained response variable, and canonical normalization.

2 The Three-Dimensional Prototype

The usual custodial picture can be read from the Hopf fibration

$$S^1 \longrightarrow S^3 \longrightarrow \mathbb{CP}^1.$$

For the Standard Model Higgs,

$$H \in \mathbb{C}^2 \simeq \mathbb{R}^4, \quad |H| = \frac{v}{\sqrt{2}},$$

the fixed-radius vacuum orbit is S^3 . The group acting on the ambient real four-space is

$$SO(4) \simeq \frac{SU(2)_L \times SU(2)_R}{\mathbb{Z}_2}.$$

The diagonal subgroup after the vacuum choice is the custodial group.

The mass relation uses the ambient quadratic form on $\mathbb{R}^4$, not only the topology of the orbit. With a single doublet vacuum

$$H_0 = \begin{pmatrix} 0 \\ v/\sqrt{2} \end{pmatrix},$$

the charged and neutral stiffnesses match:

$$\kappa_c = \kappa_n, \quad \rho = \frac{m_W^2}{m_Z^2 \cos^2 \theta_W} = 1.$$

3 The Seven-Dimensional Carrier

The CP2/Aloff–Wallach carrier is

$$SO(3) \longrightarrow X_{1,1} = SU(3)/U(1)_{1,1} \longrightarrow \mathbb{CP}^2,$$

where

$$U(1)_{1,1} = \{\text{diag}(z, z, z^{-2})\}.$$

Equivalently, in the Wilking description,

$$X_{1,1} \simeq (SU(3) \times SO(3))/U(2)_\Delta.$$

At the base point the tangent representation splits as

$$\mathfrak{m} = \mathfrak{m}_0 \oplus \mathfrak{m}_3,$$

with

$$\mathfrak{m}_0 \simeq \mathfrak{so}(3), \quad \dim_{\mathbb{R}} \mathfrak{m}_0 = 3,$$

and

$$\mathfrak{m}_3 \simeq \mathbb{C}^2 \simeq \mathbb{R}^4, \quad \dim_{\mathbb{R}} \mathfrak{m}_3 = 4.$$

The invariant metric has two scales,

$$g = x_1(-B)|_{\mathfrak{m}_0} + x_2(-B)|_{\mathfrak{m}_3}.$$

The vertical scale x_1 controls the $SO(3)$ fibre. The horizontal scale x_2 controls the Higgs four-block.

4 Why $\mathfrak{m}_3$ Is the Doublet

The complexified $\mathfrak{m}_3$ representation has the charged split

$$(\mathfrak{m}_3)_{\mathbb{C}} \simeq \mathbf{2}_{+3} \oplus \mathbf{2}_{-3}.$$

With the electroweak normalization

$$Y = \frac{q_8}{6},$$

the summand $\mathbf{2}_{+3}$ has hypercharge $Y = +1/2$. Hence the retained Higgs field is

$$H \in \mathbf{2}_{+3} \simeq \mathbb{C}_{1/2}^2.$$

The conjugate doublet belongs to $\mathbf{2}_{-3}$:

$$\tilde{H} = i\sigma_2 H^* \in \mathbb{C}_{-1/2}^2.$$

In a gauge-Higgs reading, use a higher electroweak carrier $SU(3)_W$ and embed the same $U(1)_8$. The centralizer is

$$C_{SU(3)_W}(U(1)_8) = U(2)_W.$$

The adjoint decomposes as

$$\mathfrak{su}(3)_W = (\mathbf{3}_0 \oplus \mathbf{1}_0) \oplus (\mathbf{2}_{+3} \oplus \mathbf{2}_{-3}).$$

The internal connection components in the broken $SU(3)_W/U(2)_W$ directions match the charged $\mathfrak{m}_3$ block and give one electroweak doublet.

5 The Mass-Matrix Calculation

Keep the invariant metric factor on $\mathfrak{m}_3$:

$$g|_{\mathfrak{m}_3} = x_2(-B)|_{\mathfrak{m}_3}.$$

The Higgs kinetic term has the form

$$\mathcal{L}_H = \kappa g_{\mathfrak{m}_3}(D_\mu H, D^\mu H) - V(H, \Phi_0, \dots).$$

Use the usual electroweak covariant derivative

$$D_\mu H = \partial_\mu H - igW_\mu^a \frac{\sigma_a}{2} H - ig' B_\mu \frac{1}{2} H.$$

At

$$H_0 = \begin{pmatrix} 0 \\ v/\sqrt{2} \end{pmatrix},$$

the charged terms give

$$m_W^2 = \frac{\kappa x_2 g^2 v^2}{4}$$

up to the common generator-normalization convention used for $(-B)$.

The neutral quadratic form is

$$\frac{\kappa x_2 v^2}{8} \begin{pmatrix} W_\mu^3 & B_\mu \end{pmatrix} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix} \begin{pmatrix} W^{3\mu} \\ B^\mu \end{pmatrix}.$$

The zero eigenvector is the photon. The massive eigenvalue gives

$$m_Z^2 = \frac{\kappa x_2 (g^2 + g'^2) v^2}{4}.$$

Since

$$\cos^2 \theta_W = \frac{g^2}{g^2 + g'^2},$$

the common internal factor cancels:

$$\rho = \frac{m_W^2}{m_Z^2 \cos^2 \theta_W} = 1.$$

Equivalently, in stiffness notation,

$$\kappa_c = \kappa_n.$$

In the local normalization used elsewhere in the project, the common value can be written as

$$\kappa_c = \kappa_n = \frac{3\kappa v^2 x_2}{4},$$

with the same conclusion for ρ .

6 Schur's Lemma as the Geometric Reason

The mass calculation above is the familiar one-doublet computation. The geometric reason it survives the Aloff–Wallach carrier is Schur's lemma. The block $\mathfrak{m}_3$ is one irreducible real $U(2)$ -module. Hence every $U(2)$ -invariant symmetric bilinear form on $\mathfrak{m}_3$ is a scalar multiple of the standard one:

$$g|_{\mathfrak{m}_3} = x_2(-B)|_{\mathfrak{m}_3}.$$

No independent charged and neutral stiffness can appear inside the same single $\mathfrak{m}_3$ block.

The vertical block $\mathfrak{m}_0$ has its own scale x_1 . That scale affects the fibre geometry, the squash/radion action, and KK tower data. It does not split the weak stiffnesses as long as the electroweak vacuum stays in $\mathfrak{m}_3$.

7 The Vertical Triplet Hazard

The vertical sector can contain an adjoint-like scalar Φ_0 associated with $\mathfrak{m}_0$. A minimal effective potential has the schematic form

$$V(H, \Phi_0) = m_3^2 H^\dagger H + \lambda_3 (H^\dagger H)^2 \frac{m_0^2}{2} \text{Tr}(\Phi_0^2) \lambda_{03} H^\dagger H \text{Tr}(\Phi_0^2) + \dots$$

The custodial branch uses

$$\langle H \rangle \neq 0, \quad \langle \Phi_0 \rangle = 0.$$

After the doublet condenses, stability of the vertical triplet requires

$$m_{0,\text{eff}}^2 + \frac{\lambda_{03} v^2}{2} > 0.$$

A vertical triplet vacuum gives a different vector-mass pattern and spoils the single-doublet $\rho = 1$ argument. The triplet therefore supplies a real dynamical test, not a harmless decoration.

8 D6 and D7 Roles

The D7 endpoint is $X_{1,1}$. It also fibres over the D6 flag phase:

$$S^1 = T^2/U(1)_{1,1} \longrightarrow X_{1,1} \longrightarrow Y_6 = SU(3)/T^2.$$

The D6 phase has

$$S^2 \longrightarrow Y_6 \longrightarrow \mathbb{CP}^2.$$

It carries the branch-cover and spin- c diagnostic data used elsewhere in the programme, including the branched covering

$$\mathbb{CP}^2 \longrightarrow S^4$$

with branch locus $\mathbb{RP}^2$.

For the custodial lecture, the role of D6 is branch selection. D6 defect, boundary, or CSDR/Yang–Mills terms may shift

$$m_{3,\text{eff}}^2, \quad m_{0,\text{eff}}^2, \quad \lambda_3, \quad \lambda_{03}.$$

Those terms help only if they select the doublet branch and preserve the isotropic real metric on $\mathfrak{m}_3 \simeq \mathbb{R}^4$.

The D7 gravitational action controls the radion and the squash ratio

$$a = \frac{x_1}{x_2}.$$

It can stabilize or destabilize the compactification shape. The tree-level $\rho = 1$ calculation depends on the single scalar x_2 on $\mathfrak{m}_3$, so a enters the custodial relation only through possible changes in the effective potential that selects the vacuum.

9 Comparison with $\mathbb{CP}^2 \times S^3$

The product geometry

$$\mathbb{CP}^2 \times S^3$$

is a useful comparison object. The round S^3 can model the fixed-radius orbit of a doublet. The product, however, has separate $\mathbb{CP}^2$ and S^3 factors, while $X_{1,1}$ ties the $SO(3)$ fibre and $\mathbb{CP}^2$ base through the homogeneous Aloff–Wallach structure. The split

$$\mathfrak{m} = \mathfrak{m}_0 \oplus \mathfrak{m}_3$$

with $\mathfrak{m}_3 \simeq \mathbb{C}^2$ provides the $\mathbb{CP}^2$ -indexed doublet block used in the custodial argument.

10 Brane Scan, Division Algebras, and the D6 Clue

The relevant memory is the brane-scan and division-algebra pattern, not a literal triality statement in both six and ten dimensions. Super Yang–Mills and the Green–Schwarz superstring single out

$$D = 3, 4, 6, 10,$$

corresponding to the normed division algebras

$$\mathbb{R}, \quad \mathbb{C}, \quad \mathbb{H}, \quad \mathbb{O}$$

of dimensions

$$1, \quad 2, \quad 4, \quad 8.$$

In Evans’ formulation, division algebras explain the auxiliary-field structures for super Yang–Mills in $d = 3, 4, 6, 10$.¹ Baez and Huerta give a modern self-contained account: nonabelian Yang–Mills fields minimally coupled to massless spinors are supersymmetric precisely in these spacetime dimensions, because the corresponding trilinear spinor identity is controlled by the normed division algebras.²

Triality belongs most sharply to the octonionic corner. In ten-dimensional massless physics the little group is $\text{Spin}(8)$, where the vector and two chiral spinor representations

$$\mathbf{8}_v, \quad \mathbf{8}_s, \quad \mathbf{8}_c$$

are permuted. Six dimensions occupy the quaternionic corner: the chiral spinors are quaternionic, and the Fierz identities have the same division-algebra origin, without the full $\text{Spin}(8)$ triality.

For the current programme, the useful consequence is a D6 consistency clue. A six-dimensional parent action with supersymmetric Yang–Mills ancestry can fix relative bosonic coefficients by the

¹J. M. Evans, *Auxiliary Fields for Super Yang–Mills from Division Algebras*, arXiv:[hep-th/9410239](https://arxiv.org/abs/hep-th/9410239).

²J. C. Baez and J. Huerta, *Division Algebras and Supersymmetry I*, arXiv:[0909.0551](https://arxiv.org/abs/0909.0551).

same algebraic identities that make the D6 theory consistent. Such a source action may select the $\mathfrak{m}_3$ doublet branch, suppress the $\mathfrak{m}_0$ triplet, or rule out a proposed D6 defect term. It still has to compute any De Vries candidate data from the same reduced action:

$$\mu_r, \quad \gamma(u)^2, \quad \alpha, \beta, \quad c = \frac{\beta}{\alpha^2}, \quad \sigma_r^{\text{can}}.$$

The division-algebra structure can enter as a source-action constraint. It cannot replace the source-action calculation.

11 Separation from the De Vries Problem

The custodial result proven here is

$$\mathfrak{m}_3 \simeq \mathbb{C}^2 \simeq \mathbb{R}^4, \quad \langle H \rangle \in \mathfrak{m}_3, \quad \kappa_c = \kappa_n, \quad \rho = 1.$$

The active De Vries goal asks for a different object: a source action that computes a named operator, its spectral multiplier, the response kinetic scale, and the signed Hessian coefficient in canonical variables. The custodial calculation supplies a stable electroweak control condition. It does not authorize inserting the desired spectral formula as an input.

12 Checklist for Use in a Bigger KK Paper

A larger paper can use this lecture as the custodial chapter once the following items are stated explicitly:

1. The geometry is $SO(3) \rightarrow X_{1,1} \rightarrow \mathbb{CP}^2$ with $X_{1,1} = SU(3)/U(1)_{1,1}$.
2. The tangent representation splits as $\mathfrak{m} = \mathfrak{m}_0 \oplus \mathfrak{m}_3$, with $\dim \mathfrak{m}_0 = 3$ and $\dim \mathfrak{m}_3 = 4$.
3. The electroweak doublet is the $\mathbf{2}_{+3}$ summand of $(\mathfrak{m}_3)_{\mathbb{C}}$, with $Y = q_8/6$.
4. The vacuum lies in the single $\mathfrak{m}_3$ doublet branch.
5. The vertical triplet satisfies $\langle \Phi_0 \rangle = 0$ and a positive effective mass condition.
6. Endpoint, defect, and boundary terms preserve the isotropic quadratic form on $\mathfrak{m}_3$, or their correction to ρ is computed.
7. De Vries material appears only after a source-action computation supplies the operator scale and Hessian coefficient.

13 Local Source Map

The following project files contain the longer derivations and audits behind this lecture:

- `out/CUSTODIAL_3D_TO_7D_EXTENSION.md`: Hopf-to-Aloff–Wallach custodial dictionary.
- `out/KK_HIGGSING_CUSTODIAL_MODEL.md`: gauge-Higgs model with $SU(3)_W \rightarrow U(2)_W$ and the matched $\mathfrak{m}_3$ doublet.

- `out/KK_HIGGS_POTENTIALS_1D3D_5D7D.md`: D6 and D7 potential terms, branch conditions, and product comparison.
- `out/GRAVITY_ACTION_D765_D321.md`: KK metric and Einstein–Hilbert decomposition across D7/D6/D5 and D3/D2/D1.
- `notes/Physics_First_KK_Lagrangian_Hessian_Audit.md`: separation between ordinary KK Higgs physics and the De Vries source-action problem.